Affine-Body Motion Notes

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1 Affine Motion: Definition

An *affine body* is one whose configuration at time t is described by an affine map

$$\phi(X,t) = A(t)(X - \hat{X}) + \hat{x}(t),$$

where

- $A(t) \in \mathbb{R}^{2\times 2}$ is a time-dependent linear transformation,
- $\hat{X}$ is a chosen reference point in material space,
- $\hat{x}(t)$ is the image of $\hat{X}$, interpreted as a base point.

The inverse map is

$$\phi^{-1}(x,t) = \hat{X} + A(t)^{-1}(x - \hat{x}(t)).$$

2 Velocities

2.1 Material Velocity

$$V(X,t) = \dot{A}(t)(X - \hat{X}) + \hat{v}(t), \qquad \hat{v}(t) = \dot{\hat{x}}(t).$$

2.2 Spatial Velocity

For $x = \phi(X,t)$,

$$v(x,t) = B(t)(x - \hat{x}(t)) + \hat{v}(t), \qquad B(t) = \dot{A}(t)A(t)^{-1}.$$

3 Linear Momentum and Center of Mass

For a system of particles (m_i, v_i),

$$p = \sum_i m_i v_i, \qquad m = \sum_i m_i, \qquad v_{\text{com}} = \frac{p}{m}.$$

In affine motion,

$$v_i = B(x_i - \hat{x}) + \hat{v}.$$

Thus,

$$p = B\sum_i m_i(x_i - \hat{x}) + m\hat{v}.$$

If $\hat{x} = x_{\text{com}} = \frac{1}{m}\sum_i m_i x_i$, then

$$\sum_i m_i(x_i - \hat{x}) = \sum_i m_i x_i - m\hat{x} = m x_{\text{com}} - m x_{\text{com}} = 0.$$

Hence the first term vanishes, leaving

$$p = m\hat{v} \quad \Rightarrow \quad \hat{v} = \frac{p}{m} = v_{\text{com}}.$$

Thus the base velocity coincides with the center-of-mass velocity.

4 Angular Momentum

The angular momentum about $\hat{x}$ is

$$l = \sum_i (x_i - \hat{x}) \times p_i, \qquad p_i = m_i v_i.$$

For vectors $a = (a_1, a_2)$ and $b = (b_1, b_2)$ in $\mathbb{R}^2$, the scalar cross product is

$$a \times b = a_1 b_2 - a_2 b_1.$$

This can be written using the 90° rotation matrix

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad a \times b = a \cdot (Jb).$$

Let $r_i = x_i - \hat{x}$. Then

$$r_i \times p_i = r_i \cdot (Jp_i) = -p_i \cdot (Jr_i).$$

Define the rotated relative position

$$\tilde{u}_i := Jr_i = (x_i - \hat{x}) \times k,$$

where k is the out-of-plane unit vector. Hence

$$r_i \times p_i = -p_i \cdot \tilde{u}_i.$$

Summing over all particles,

$$l = \sum_i (x_i - \hat{x}) \times p_i = -\sum_i p_i \cdot \tilde{u}_i.$$

5 Mathematical Analysis of the Affine Velocity Field

We consider three points

$$x_1 = \left(0, \tfrac{1}{2}\right), \quad x_2 = \left(0, -\tfrac{1}{2}\right), \quad x_3 = (1, 0),$$

with centroid

$$\hat{x} = \left(\tfrac{1}{3}, 0\right),$$

and prescribed velocities

$$v(x_1) = v(x_2) = (0, 0), \qquad v(x_3) = (\varepsilon, 0), \quad \varepsilon < 0.$$

Affine Ansatz

We approximate the velocity field using the six affine basis modes

$$v(x) \;=\; \sum_{k=1}^{6} c_k\, \tilde{u}^{(k)}(x),$$

where

$$\tilde{u}^{(k)}(x) = \begin{cases} A^{(k)}(x-\hat{x}), & k = 1,\dots,4, \\ (1,0), & k = 5, \\ (0,1), & k = 6, \end{cases}$$

and the linear modes are

$$A^{(1)} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A^{(2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad A^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad A^{(4)} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Forming the 6×6 System

At each node x_i we require

$$v(x_i) \;=\; \sum_{k=1}^{6} c_k \tilde{u}^{(k)}(x_i).$$

Stacking the x– and y–components for the three points gives the 6×6 linear system

$$\underbrace{\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & -\frac{1}{3} & 0 & 1 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & 0 & \frac{1}{2} & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{3} & 0 & 1 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & 0 & -\frac{1}{2} & 0 & 1 \\ 0 & 0 & \frac{2}{3} & 0 & 1 & 0 \\ \frac{2}{3} & \frac{2}{3} & 0 & 0 & 0 & 1 \end{bmatrix}}_{G} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \varepsilon \\ 0 \end{bmatrix}}_{b}.$$

Solving the System

Row reductions yield

$$c_1 = c_2 = c_4 = c_6 = 0, \qquad c_3 = \varepsilon, \qquad c_5 = \tfrac{\varepsilon}{3}.$$

Resulting Velocity Field

Thus

$$v(x) \;=\; \varepsilon\, A^{(3)}(x-\hat{x}) \;+\; \tfrac{\varepsilon}{3}(1,0) \;=\; \boxed{(\varepsilon x,\, 0)}.$$

Flow Map and Interpretation

The trajectories follow

$$\dot{x} = \varepsilon x, \qquad \dot{y} = 0,$$

so that for (x_0, y_0),

$$x(t) = x_0 e^{\varepsilon t}, \qquad y(t) = y_0.$$

Hence x_1 and x_2 remain fixed, while

$$x_3(t) = (e^{\varepsilon t}, 0)$$

slides toward the line through x_1 and x_2. For $\varepsilon < 0$, the approach is exponential: fast at first, then slowing asymptotically, preventing collision in finite time.

6 2D Affine-Body Dynamics Test

We summarize the procedure for testing affine-body dynamics using a fixed reference configuration and a moving segment. The formulation follows the affine projection method, with Schur decomposition enabling exact time integration.

6.1 Problem Setup and Particle Model

We represent particles in the plane by samples

$$S_i = \{x_i, v_i, m_i\}, \quad i = 1, \ldots, N,$$

with position $x_i \in \mathbb{R}^2$, velocity $v_i \in \mathbb{R}^2$, and mass $m_i > 0$.

The center of mass (COM) is

$$x_{\text{COM}} = \frac{1}{M} \sum_{i=1}^{N} m_i x_i, \qquad M = \sum_{i=1}^{N} m_i.$$

For testing, we consider $N = 5$ points:

- Two moving points forming a segment along the x-axis, with initial upward velocity.
- Three fixed points forming a reference segment along $y = 2$, with zero velocity.

6.2 Affine Projection onto Velocity Basis

To represent particle velocities in a reduced form, we project them onto the affine velocity space spanned by six basis functions $\{u^{(k)}\}_{k=1}^{6}$:

$$u(x) = \sum_{k=1}^{6} c_k \, u^{(k)}(x),$$

where the basis functions at a particle position x (relative to the COM x_{COM}) are

$$\begin{aligned}
u^{(1)}(x) &= A_1(x - x_{\text{COM}}), & A_1 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \\
u^{(2)}(x) &= A_2(x - x_{\text{COM}}), & A_2 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \\
u^{(3)}(x) &= A_3(x - x_{\text{COM}}), & A_3 &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \\
u^{(4)}(x) &= A_4(x - x_{\text{COM}}), & A_4 &= \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \\
u^{(5)}(x) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad u^{(6)}(x) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.
\end{aligned}$$

The coefficients $c = (c_1, \ldots, c_6)^T$ are determined by the mass inner-product projection condition:

$$\langle u, u^{(j)} \rangle_m = \langle v, u^{(j)} \rangle_m, \qquad j = 1, \ldots, 6,$$

where

$$\langle f, g \rangle_m = \sum_{i=1}^{N} m_i f(x_i) \cdot g(x_i).$$

This leads to the symmetric linear system

$$Gc = b,$$

with entries

$$G_{jk} = \sum_{i=1}^{N} m_i \, u^{(j)}(x_i) \cdot u^{(k)}(x_i), \qquad b_j = \sum_{i=1}^{N} m_i \, u^{(j)}(x_i) \cdot v_i.$$

6.3 From Projection Coefficients to Affine Field

Once the coefficients c are determined, they define the affine velocity field. Specifically,

$$B = c_1 A_1 + c_2 A_2 + c_3 A_3 + c_4 A_4, \qquad \hat{v} = \begin{bmatrix} c_5 \\ c_6 \end{bmatrix},$$

so that

$$\frac{\partial x}{\partial t} = B(x - x_{\mathrm{COM}}) + \hat{v}.$$

To integrate the affine dynamics exactly, we transform the linear part B into Schur form. Let

$$B = QTQ^T, \qquad h(t) = Q^T\big(x(t) - x_{\mathrm{COM}}\big),$$

where Q is orthogonal and T is a 2×2 real Schur block. In these coordinates, the ODE becomes

$$\dot{h}(t) = Th(t) + v_Q, \qquad v_Q = Q^T \hat{v}.$$

Depending on the spectrum of B, T takes one of two forms:

- a real upper-triangular block $T = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$, corresponding to two real eigenvalues, or
- a 2×2 rotation–scaling block $T = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$, corresponding to a complex-conjugate pair $a \pm ib$.

In either case, the system in Schur coordinates admits the exact update

$$h(t + \Delta t) = e^{T\Delta t} h(t) + T^{-1}\big(e^{T\Delta t} - I\big) v_Q.$$

Returning to original variables,

$$x(t + \Delta t) = Qe^{T\Delta t} Q^T (x(t) - x_{\mathrm{COM}}) + QT^{-1}(e^{T\Delta t} - I) Q^T \hat{v} + x_{\mathrm{COM}}.$$

Thus the task reduces to evaluating $e^{T\Delta t}$ and $T^{-1}(e^{T\Delta t} - I)$ explicitly, which we derive case by case.

6.4 Exact One-Step Update: Real Schur Block

We consider the ODE in Schur coordinates:

$$h'(t) = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} h(t) + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad h(t) = \begin{bmatrix} h_1(t) \\ h_2(t) \end{bmatrix}.$$

We seek the exact update from time t to $t + \Delta t$, with $\Delta t > 0$.

Homogeneous solution. For the homogeneous system $h'(t) = Th(t)$:

- Second component:
$$h_2^{\text{hom}}(t + \Delta t) = e^{d\Delta t}\, h_2(t).$$

- First component:
$$h_1'(s) - ah_1(s) = be^{d(s-t)}h_2(t), \quad s \in [t, t + \Delta t].$$
Multiplying by $e^{-a(s-t)}$ and integrating:
$$h_1^{\text{hom}}(t + \Delta t) = e^{a\Delta t}h_1(t) + \frac{b}{d-a}\Big(e^{d\Delta t} - e^{a\Delta t}\Big)h_2(t).$$

- Thus the homogeneous step is
$$h^{\text{hom}}(t + \Delta t) = e^{T\Delta t}h(t), \qquad e^{T\Delta t} = \begin{bmatrix} e^{a\Delta t} & \frac{b}{d-a}(e^{d\Delta t} - e^{a\Delta t}) \\ 0 & e^{d\Delta t} \end{bmatrix}.$$

Inhomogeneous contribution. By variation of constants,

$$h(t + \Delta t) = e^{T\Delta t}h(t) + \int_0^{\Delta t} e^{Ts}\tilde{v}\, ds, \quad \tilde{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

Since $\frac{d}{ds}e^{Ts} = Te^{Ts}$, we have

$$\int_0^{\Delta t} e^{Ts}\, ds = T^{-1}\big(e^{T\Delta t} - I\big),$$

and hence

$$\int_0^{\Delta t} e^{Ts}\tilde{v}\, ds = T^{-1}\big(e^{T\Delta t} - I\big)\tilde{v}.$$

The inverse of T is

$$T^{-1} = \begin{bmatrix} 1/a & -b/(ad) \\ 0 & 1/d \end{bmatrix},$$

and

$$e^{T\Delta t} - I = \begin{bmatrix} e^{a\Delta t} - 1 & \frac{b}{d-a}(e^{d\Delta t} - e^{a\Delta t}) \\ 0 & e^{d\Delta t} - 1 \end{bmatrix}.$$

Thus

$$T^{-1}(e^{T\Delta t} - I) = \begin{bmatrix} \frac{e^{a\Delta t}-1}{a} & \frac{b}{a(d-a)}\big(e^{d\Delta t} - e^{a\Delta t}\big) - \frac{b}{ad}(e^{d\Delta t} - 1) \\ 0 & \frac{e^{d\Delta t}-1}{d} \end{bmatrix}.$$

Final update. The exact step from t to $t+\Delta t$ is

$$h(t+\Delta t) = e^{T\Delta t}h(t) + T^{-1}(e^{T\Delta t} - I)\tilde{v}.$$

In components:

$$\boxed{\begin{aligned} h_2(t+\Delta t) &= e^{d\Delta t}\, h_2(t) + \frac{e^{d\Delta t}-1}{d}\, v_2, \\ h_1(t+\Delta t) &= e^{a\Delta t}\, h_1(t) + \frac{b}{d-a}\Big(e^{d\Delta t} - e^{a\Delta t}\Big) h_2(t) \\ &\qquad + \frac{e^{a\Delta t}-1}{a}\, v_1 + \left[\frac{b}{a(d-a)}\Big(e^{d\Delta t} - e^{a\Delta t}\Big) - \frac{b}{ad}\Big(e^{d\Delta t}-1\Big)\right] v_2. \end{aligned}}$$

Degenerate limits. The above expressions remain well defined in the following limits:

- If $a = 0$, use $\frac{e^{a\Delta t}-1}{a} \to \Delta t$.
- If $d = 0$, use $\frac{e^{d\Delta t}-1}{d} \to \Delta t$.
- If $a = d$, use $\frac{e^{d\Delta t}-e^{a\Delta t}}{d-a} \to \Delta t e^{a\Delta t}$.

Thus the solutions in degenerate cases become:

$$\begin{aligned} &h_2(t+\Delta t) = h_2(t) + \Delta t\, v_2, && (d=0), \\ &h_1(t+\Delta t) = h_1(t) + b\Delta t\, h_2(t) + \Delta t\, v_1 - \tfrac{b}{a}\big(\Delta t - \tfrac{e^{a\Delta t}-1}{a}\big) v_2, && (a=0), \\ &h_1(t+\Delta t) = e^{a\Delta t}\Big(h_1(t) + b\Delta t\, h_2(t)\Big) + \tfrac{e^{a\Delta t}-1}{a}\, v_1 + \big[\tfrac{b}{a^2}\big(e^{a\Delta t} - 1 - a\Delta t\big)\big]\, v_2, && (a=d). \end{aligned}$$

These cover all singular limits, ensuring the update rule is valid for all (a, d).

6.5 Exact One-Step Update: Complex Schur Block

When B has a complex-conjugate pair of eigenvalues, its *real* Schur form contains a 2×2 quasi-triangular block of the form

$$T \;=\; \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \;=\; aI + bJ, \qquad J \;=\; \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad J^2 = -I,$$

whose eigenvalues are $a \pm ib$. In Schur coordinates the ODE is

$$\dot{h}(t) \;=\; T\, h(t) + v_Q, \qquad h(t) = \begin{bmatrix} h_1(t) \\ h_2(t) \end{bmatrix}, \quad v_Q = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

We derive the exact one-step update from t to $t+\Delta t$.

Homogeneous solution. Using $T = aI + bJ$ with $J^2 = -I$,

$$e^{Ts} = e^{as}\big(\cos(bs)\, I + \sin(bs)\, J\big) = e^{as}\begin{bmatrix} \cos(bs) & \sin(bs) \\ -\sin(bs) & \cos(bs) \end{bmatrix}.$$

Therefore, with $E \equiv e^{a\Delta t}$, $c \equiv \cos(b\Delta t)$, $s \equiv \sin(b\Delta t)$,

$$h^{\text{hom}}(t+\Delta t) = e^{T\Delta t}h(t) = E\begin{bmatrix} c & s \\ -s & c \end{bmatrix}\begin{bmatrix} h_1(t) \\ h_2(t) \end{bmatrix}.$$

Inhomogeneous contribution (variation of constants). As usual,

$$h(t+\Delta t) \;=\; e^{T\Delta t}h(t) \;+\; \int_0^{\Delta t} e^{Ts}\, v_Q\, ds \;=\; e^{T\Delta t}h(t) \;+\; T^{-1}\big(e^{T\Delta t}-I\big)v_Q,$$

because $\frac{d}{ds}e^{Ts} = Te^{Ts}$. We now compute $e^{T\Delta t} - I$ and multiply by T^{-1}.

First,

$$e^{T\Delta t} - I = E\big(c\,I + s\,J\big) - I = A\,I + B\,J, \qquad A \equiv Ec - 1, \quad B \equiv Es.$$

Next, since

$$T^{-1} = \frac{1}{a^2+b^2}\,(aI - bJ) = \frac{1}{a^2+b^2}\begin{bmatrix} a & -b \\ b & a \end{bmatrix},$$

we get

$$T^{-1}\big(e^{T\Delta t} - I\big) = \frac{1}{a^2+b^2}\Big((aA+bB)\,I + (aB-bA)\,J\Big) = \frac{1}{a^2+b^2}\begin{bmatrix} aA+bB & aB-bA \\ -(aB-bA) & aA+bB \end{bmatrix}.$$

Final exact update (matrix form). Collecting terms,

$$\boxed{h(t+\Delta t) \;=\; R\,h(t) \;+\; F\,v_Q, \qquad R \;=\; E\begin{bmatrix} c & s \\ -s & c \end{bmatrix}, \quad F \;=\; \frac{1}{a^2+b^2}\begin{bmatrix} aA+bB & aB-bA \\ -(aB-bA) & aA+bB \end{bmatrix},}$$

with $E = e^{a\Delta t}$, $c = \cos(b\Delta t)$, $s = \sin(b\Delta t)$, $A = Ec - 1$, and $B = Es$.

Final exact update (component form).

$$h_1(t+\Delta t) = E\big(c\,h_1(t) + s\,h_2(t)\big) + \frac{(aA+bB)\,v_1 + (aB-bA)\,v_2}{a^2+b^2},$$

$$h_2(t+\Delta t) = E\big(-s\,h_1(t) + c\,h_2(t)\big) + \frac{-(aB-bA)\,v_1 + (aA+bB)\,v_2}{a^2+b^2}.$$

Degenerate limits and numerical robustness.

- $b \to 0$ (block $\to aI$): $R \to e^{a\Delta t}I$ and $F \to \frac{e^{a\Delta t}-1}{a}\,I$; if also $a \to 0$, then $F \to \Delta t\, I$.
- $a = 0$ (pure rotation): $E = 1$, $A = c - 1$, $B = s$; thus

$$F = \begin{bmatrix} s/b & -(c-1)/b \\ (c-1)/b & s/b \end{bmatrix} \xrightarrow[b\to 0]{} \Delta t\, I.$$

- Small $|a|\Delta t$ and/or $|b|\Delta t$: to avoid cancellation, one may use the series

$$R = I + T\Delta t + \tfrac{1}{2}T^2\Delta t^2 + O(\Delta t^3), \qquad F = \Delta t\Big(I + \tfrac{1}{2}T\Delta t + \tfrac{1}{6}T^2\Delta t^2 + \cdots\Big),$$

  with $T^2 = (a^2 - b^2)I + 2ab\,J$.

Back to original variables. As before, $h = Q^T(x - x_{\mathrm{COM}})$ and $v_Q = Q^T\hat{v}$, hence

$$x(t+\Delta t) = Q\big(R\,h(t) + F\,v_Q\big) + x_{\mathrm{COM}} = QRQ^T\big(x(t) - x_{\mathrm{COM}}\big) + QFQ^T\hat{v} + x_{\mathrm{COM}}.$$

6.6 Unified Summary

For both Schur block structures, the exact one-step update in Schur coordinates has the form

$$h(t+\Delta t) = e^{T\Delta t}h(t) + T^{-1}\big(e^{T\Delta t} - I\big)v_Q,$$

with closed expressions depending on the block:

- *Real-eigenvalue triangular block*: explicit update derived in the real case.
- *Complex-eigenvalue block*: explicit update derived in the conjugate-pair case.

Both cases admit continuous degenerate limits, ensuring numerical robustness.

7 Incremental Potential Minimization with Spring and Barrier Forces

At each time step, the updated configuration $\mathbf{x}^{n+1}$ is obtained by minimizing the incremental potential energy

$$\Phi(\mathbf{x}) = \tfrac{1}{2}\,\|\mathbf{x} - \hat{\mathbf{x}}\|_M^2 + (\Delta t)^2\, PE(\mathbf{x}), \qquad \|\mathbf{v}\|_M^2 = \mathbf{v}^\mathsf{T} M \mathbf{v},$$

where $\hat{\mathbf{x}} = \mathbf{x}^n + \Delta t\, \mathbf{v}^n$ is the explicit predictor, and $M = \mathrm{diag}(m_1 I, \ldots, m_N I)$ is the lumped mass matrix with 2×2 identity blocks in 2D. The total potential energy is

$$PE(\mathbf{x}) = PE^{\mathrm{s}}(\mathbf{x}) + PE^{\mathrm{g}}(\mathbf{x}) + PE^{\mathrm{bar}}(\mathbf{x}),$$

including the spring stretching energy, gravitational potential, and node–segment barrier potential.

7.1 Gravitational Potential

Each node i of mass m_i experiences a constant gravitational acceleration $\mathbf{g}$:

$$PE_i^{\mathrm{g}}(\mathbf{x}_i) = -\, m_i\, \mathbf{g}^\mathsf{T}\mathbf{x}_i, \qquad \nabla_{\mathbf{x}_i} PE_i^{\mathrm{g}} = -\, m_i\, \mathbf{g}, \qquad \nabla^2_{\mathbf{x}_i\mathbf{x}_i} PE_i^{\mathrm{g}} = 0.$$

7.2 Spring Potential Energy

Each edge $(i,j) \in \mathcal{E}$ behaves as a linear spring with stiffness k and rest length $L_{ij} = \|\mathbf{X}_i - \mathbf{X}_j\|$. Let $\ell_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ denote its current length. Then

$$PE^{\mathrm{s}}(\mathbf{x}) = \sum_{(i,j)\in\mathcal{E}} \frac{k}{2L_{ij}}\,(\ell_{ij} - L_{ij})^2.$$

Gradient. For one spring with $\mathbf{r} = \mathbf{x}_i - \mathbf{x}_j$ and $\ell = \|\mathbf{r}\|$,

$$E_{ij}^{\mathrm{s}} = \frac{k}{2L_{ij}}(\ell - L_{ij})^2, \qquad \nabla_{\mathbf{x}_i} E_{ij}^{\mathrm{s}} = \frac{k}{L_{ij}}(\ell - L_{ij})\,\frac{\mathbf{r}}{\ell}, \qquad \nabla_{\mathbf{x}_j} E_{ij}^{\mathrm{s}} = -\,\nabla_{\mathbf{x}_i} E_{ij}^{\mathrm{s}}.$$

Hessian. Differentiating $\nabla_{\mathbf{x}_i} E_{ij}^{\mathrm{s}}$ with respect to $\mathbf{x}_i$, let $\alpha = \frac{k}{L_{ij}}(\ell - L_{ij})$ and $\mathbf{u} = \mathbf{r}/\ell$. We obtain

$$K_{ij} = \frac{k}{L_{ij}}\left[\mathbf{u}\mathbf{u}^\top + \frac{\ell - L_{ij}}{\ell}(I - \mathbf{u}\mathbf{u}^\top)\right].$$

7.3 C^1 Continuity of the Point–Segment Distance

Let segment endpoints be $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^2$, with

$$\mathbf{s} = \mathbf{x}_2 - \mathbf{x}_1, \qquad L = \|\mathbf{s}\|, \qquad \mathbf{u} = \frac{\mathbf{s}}{L}.$$

For any point $\mathbf{x} \in \mathbb{R}^2$, define

$$t = \frac{(\mathbf{x} - \mathbf{x}_1)^\top \mathbf{s}}{L}, \qquad \hat{t} = \min\{L, \max\{0, t\}\}, \qquad \hat{\mathbf{x}} = \mathbf{x}_1 + \frac{\hat{t}}{L}\mathbf{s},$$

and the distance

$$d(\mathbf{x}, [\mathbf{x}_1, \mathbf{x}_2]) = \|\mathbf{x} - \hat{\mathbf{x}}\|.$$

Let $\mathbf{P} = I - \mathbf{u}\mathbf{u}^\top$ denote the orthogonal projector. Then

$$d(\mathbf{x}, [\mathbf{x}_1, \mathbf{x}_2]) = \begin{cases} \|\mathbf{P}(\mathbf{x} - \mathbf{x}_1)\|, & 0 < t < L, \\ \|\mathbf{x} - \mathbf{x}_1\|, & t \le 0, \\ \|\mathbf{x} - \mathbf{x}_2\|, & t \ge L. \end{cases}$$

Gradient continuity. Within each region,

$$\nabla d(\mathbf{x}) = \begin{cases} \dfrac{\mathbf{P}(\mathbf{x} - \mathbf{x}_1)}{\|\mathbf{P}(\mathbf{x} - \mathbf{x}_1)\|}, & 0 < t < L, \\ \dfrac{\mathbf{x} - \mathbf{x}_1}{\|\mathbf{x} - \mathbf{x}_1\|}, & t < 0, \\ \dfrac{\mathbf{x} - \mathbf{x}_2}{\|\mathbf{x} - \mathbf{x}_2\|}, & t > L. \end{cases}$$

At $t = 0$ and $t = L$, the limits coincide, hence d is C^1 but not C^2.

7.4 Node–Segment Barrier Potential

To prevent interpenetration, we introduce a C^1-continuous barrier potential between a dynamic node $\mathbf{x}_i$ and a segment $(\mathbf{x}_j, \mathbf{x}_{j+1})$. Let

$$\mathbf{s} = \mathbf{x}_{j+1} - \mathbf{x}_j, \qquad L = \|\mathbf{s}\|, \qquad \mathbf{u} = \frac{\mathbf{s}}{L}, \qquad \mathbf{P} = I - \mathbf{u}\mathbf{u}^\top.$$

For the interior case ($0 < t^* < 1$),

$$\mathbf{r}^* = \mathbf{P}(\mathbf{x}_i - \mathbf{x}_j), \qquad d^* = \|\mathbf{r}^*\|, \qquad \mathbf{n} = \frac{\mathbf{r}^*}{d^*}.$$

Barrier function. Given activation distance $\hat{d} > 0$, define

$$b(d; \hat{d}) = \begin{cases} -(d - \hat{d})^2 \ln(d/\hat{d}), & 0 < d < \hat{d}, \\ 0, & d \ge \hat{d}, \end{cases}$$

with

$$b'(d) = -2(d - \hat{d}) \ln(d/\hat{d}) - \frac{(d - \hat{d})^2}{d}, \qquad b''(d) = -2 \ln(d/\hat{d}) - \frac{4(d - \hat{d})}{d} + \frac{(d - \hat{d})^2}{d^2}.$$

Barrier energy. The barrier energy for one node–segment pair is

$$PE^{\text{bar}}_{i,j,j+1} = b(d^*; \hat{d}), \qquad d^* = \|\mathbf{P}(\mathbf{x}_i - \mathbf{x}_j)\|.$$

Gradient.

$$\nabla_{\mathbf{x}_i} d^* = \mathbf{P}\mathbf{n},$$

and by the chain rule,

$$\nabla_{\mathbf{x}_i} PE^{\text{bar}} = \mathbf{P}\, b'(d^*)\, \mathbf{n}.$$

Hessian. Starting from $\nabla_{\mathbf{x}_i} PE^{\text{bar}} = \mathbf{P}\, b'(d^*)\, \mathbf{n}$,

$$\nabla^2_{\mathbf{x}_i\mathbf{x}_i} PE^{\text{bar}} = \mathbf{P}\Big[b''(d^*)\,(\nabla_{\mathbf{x}_i} d^*)\mathbf{n}^\top + b'(d^*)\,\nabla_{\mathbf{x}_i}\mathbf{n}\Big].$$

Since $\nabla_{\mathbf{x}_i} d^* = \mathbf{P}\mathbf{n}$ and $\nabla_{\mathbf{x}_i}\mathbf{n} = \frac{1}{d^*}(I - \mathbf{n}\mathbf{n}^\top)\mathbf{P}$, we have

$$H^{\text{bar}}_{ii} = \mathbf{P}\left[b''(d^*)\,\mathbf{n}\mathbf{n}^\top + \frac{b'(d^*)}{d^*}(I - \mathbf{n}\mathbf{n}^\top)\right]\mathbf{P}.$$

At endpoints ($t^* = 0$ or $t^* = 1$), replace $\mathbf{P}$ by I for the active endpoint.

Smoothness. Each branch is analytic (C^∞), and ∇PE^{bar} is continuous across transitions at $t^* = 0, 1$. The Hessian exhibits finite jumps, hence PE^{bar} is globally C^1 but not C^2.

7.5 Gauss–Seidel Incremental Minimization

We minimize $\Phi(\mathbf{x})$ by nonlinear Gauss–Seidel iteration, updating one node at a time:

$$\nabla_{\mathbf{x}_i}\Phi(\mathbf{x}) = m_i(\mathbf{x}_i - \hat{\mathbf{x}}_i) + (\Delta t)^2\big(\nabla_{\mathbf{x}_i} PE^{\text{s}} + \nabla_{\mathbf{x}_i} PE^{\text{bar}} - m_i\mathbf{g}\big) = 0.$$

Linearizing locally,

$$D_i = m_i I + (\Delta t)^2\big(H^{\text{s}}_{ii} + H^{\text{bar}}_{ii}\big), \qquad \delta_i = -\,D_i^{-1}\mathbf{g}_i, \qquad \mathbf{x}_i \leftarrow \mathbf{x}_i + \omega\,\delta_i,$$

with relaxation $\omega \in (0, 1]$; fixed nodes are skipped.

Convergence. Sweeps continue until $\max_i \|\mathbf{g}_i\| < \varepsilon_{\text{abs}}$ or a sweep limit is reached. This yields a fully implicit, unconditionally stable integration of the mass–spring system with differentiable barrier contact.

Remark 7.1 (On Barrier Coupling and Solver Consistency)**.** *Earlier implementations omitted barrier forces from the Gauss–Seidel loop, evaluating them only afterward. Although the solver converged numerically, it ignored contact reactions during minimization. Numerical convergence does not imply physical correctness: contact forces must be evaluated within the nonlinear solve using the current geometry. Embedding barrier evaluation directly into the Gauss–Seidel iterations ensures that springs, gravity, and contact all participate in the same implicit equilibrium, preserving consistency and energy behavior.*

Remark 7.2 (Implementation Note on State Synchronization)**.** *Because each node update modifies the shared geometry, the global state must be synchronized immediately after each local update. Failing to propagate* $\mathbf{x}_i$ *from the local domain to the global configuration can cause subsequent evaluations (e.g. barrier gradients or CCD checks) to operate on stale geometry, effectively delaying contact response by one sweep. Updating both the local and global positions after every node step ensures that the barrier terms are computed with the most recent configuration, maintaining solver consistency and correct collision timing. This synchronization is essential when both chains move simultaneously, as the barrier geometry depends on all currently active nodes.*

Remark 7.3 (CCD Line Search)**.** *When combined with continuous collision detection (CCD), a per-node line search can adaptively scale* ω *to prevent penetration during large updates. This CCD weight factor acts as a physically grounded safeguard, replacing arbitrary damping by a collision-aware step limiter. In the fully dynamic case, CCD must consider the relative motion of both the point and the segment endpoints, i.e. a point–moving-segment query, ensuring guaranteed separation under simultaneous motion.*

8 2D Continuous Collision Detection

In two dimensions, the only nontrivial continuous collision detection (CCD) case occurs between a *dynamic node* and a *moving segment*. Consider a moving node $\mathbf{x}_1$ and a segment defined by moving endpoints $\mathbf{x}_2$ and $\mathbf{x}_3$. Each vertex moves linearly with its own displacement:

$$\mathbf{x}_i(t) = \mathbf{x}_i + t\,\Delta\mathbf{x}_i, \quad i = 1, 2, 3.$$

Thus,

$$\begin{aligned}
\mathbf{x}_1(t) &= \mathbf{x}_1 + t\,\Delta\mathbf{x}_1,\\
\mathbf{x}_2(t) &= \mathbf{x}_2 + t\,\Delta\mathbf{x}_2,\\
\mathbf{x}_3(t) &= \mathbf{x}_3 + t\,\Delta\mathbf{x}_3.
\end{aligned}$$

8.1 Signed Area Function

Define the signed area between the moving segment and the node as

$$f(t) = (\mathbf{x}_3(t) - \mathbf{x}_2(t)) \times (\mathbf{x}_1(t) - \mathbf{x}_2(t)).$$

This scalar represents twice the signed area of the triangle formed by the three vertices at time t. A collision occurs when $f(t) = 0$, i.e., when the three points become collinear and the node crosses the line of the moving segment.

8.2 Quadratic Formulation

Because each $\mathbf{x}_i(t)$ is linear in t, the signed area function $f(t)$ is quadratic:

$$f(t) = at^2 + bt + c,$$

with coefficients

$$\boxed{\begin{aligned}
a &= (\Delta\mathbf{x}_3 - \Delta\mathbf{x}_2) \times (\Delta\mathbf{x}_1 - \Delta\mathbf{x}_2),\\
b &= (\Delta\mathbf{x}_3 - \Delta\mathbf{x}_2) \times (\mathbf{x}_1 - \mathbf{x}_2) + (\mathbf{x}_3 - \mathbf{x}_2) \times (\Delta\mathbf{x}_1 - \Delta\mathbf{x}_2),\\
c &= (\mathbf{x}_3 - \mathbf{x}_2) \times (\mathbf{x}_1 - \mathbf{x}_2).
\end{aligned}}$$

The potential contact times satisfy $f(t) = 0$, yielding up to two roots:

$$t_{\pm} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Only real roots $t_{\pm} \in [0, 1]$ correspond to valid collision times.

8.3 Numerically Stable Quadratic Solve

The direct quadratic formula can suffer from numerical cancellation when $|b| \approx \sqrt{b^2 - 4ac}$. To obtain stable roots, define

$$D = b^2 - 4ac, \qquad s = \operatorname{sign}(b), \qquad q = -\tfrac{1}{2}\big(b + s\sqrt{D}\big).$$

Then the two roots can be computed robustly as

$$t_+ = \frac{q}{a}, \qquad t_- = \frac{c}{q}.$$

These are algebraically equivalent to the standard quadratic formula but avoid subtracting nearly equal numbers. Indeed, since $(-b + s\sqrt{D}) = -2q$, one finds

$$t_+ = \frac{-b + s\sqrt{D}}{2a} = \frac{q}{a}, \qquad t_- = \frac{c}{a\,t_+} = \frac{c}{q}.$$

This pair of formulas is numerically stable for all magnitudes of a and b.

Remark 8.1. *The quadratic equation $f(t) = 0$ can yield up to two real roots t_+ and t_-. However, only one of them corresponds to the actual collision. The function $f(t)$ measures the signed area of the triangle formed by the node and the segment; it changes sign once when the node crosses the segment line. Hence, only the first sign change within $[0, 1]$ represents a valid geometric crossing. Therefore, the collision time is chosen as the earliest valid root:*

$$t^* = \min\{\, t_{\pm} \mid t_{\pm} \in [0, 1] \,\}.$$

This t^* is the first time of contact used in continuous collision detection (CCD).

8.4 Inside-Segment Test

At the candidate time t^*, project the node onto the segment to verify that it lies between the segment endpoints:

$$s = \frac{(\mathbf{x}_1(t^*) - \mathbf{x}_2(t^*)) \cdot (\mathbf{x}_3(t^*) - \mathbf{x}_2(t^*))}{\|\mathbf{x}_3(t^*) - \mathbf{x}_2(t^*)\|^2}, \qquad 0 \le s \le 1.$$

If both $f(t^*) = 0$ and $0 \le s \le 1$ hold, the node collides with the segment at time t^*.

8.5 Degenerate Case ($a \approx 0$)

When $\Delta\mathbf{x}_3 \approx \Delta\mathbf{x}_2$, the term

$$a = (\Delta\mathbf{x}_3 - \Delta\mathbf{x}_2) \times (\Delta\mathbf{x}_1 - \Delta\mathbf{x}_2) \approx 0.$$

Then $f(t)$ degenerates to a nearly linear function in t. Let $D = b^2 - 4ac$ and $s = \text{sign}(b)$. The quadratic roots are

$$t_{\pm} = \frac{-b \pm \sqrt{D}}{2a}.$$

To examine the limit as $a \to 0$, we rewrite the stable form:

$$t_{+} = \frac{-2c}{b + s\sqrt{D}}, \qquad t_{-} = \frac{-b - s\sqrt{D}}{2a}.$$

As $\sqrt{D} \to |b|$ for small a, the finite root tends to

$$t^* = \lim_{a \to 0} t_{+} = -\frac{c}{b},$$

which matches the linear solution of $b\,t + c = 0$. The other root t_{-} diverges as $a \to 0$ and is therefore discarded.

8.6 Solver Update Rule

Once the earliest valid contact time t^* is found, the solver must restrict its update to prevent interpenetration. A safe step rule is defined as

$$\omega_{\text{safe}} = \begin{cases} 1, & \text{if } t^* > 1, \\ 0.9\,t^*, & \text{otherwise.} \end{cases}$$

The factor 0.9 introduces a small safety margin, ensuring that the updated positions remain strictly collision-free. The new configuration is then updated as

$$\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + \omega_{\text{safe}}\,\Delta\mathbf{x}_i, \quad i = 1, 2, 3.$$